

Tobias Gerstenberg

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Professional experience and affiliations

Sep 2026 - present	Associate Professor of Psychology and Computer Science (by courtesy) PI of the Causality in Cognition Lab (CICL)	Stanford, USA
Sep 2018 - Aug 2026	Assistant Professor of Psychology	Stanford, USA
Apr 2014 - Jul 2018	Postdoctoral Associate PI: Prof. Joshua Tenenbaum	MIT, USA
Apr 2013 - Apr 2014	Postdoctoral Fellow PI: Prof. Joshua Tenenbaum	MIT, USA

Education

Aug 2009 - Feb 2013	PhD Candidate Advisor: Prof. David Lagnado	University College London, England
Oct 2008 - Aug 2009	Course Attendance Psychology, Computer Science, Philosophy, and Theology	Humboldt University, Berlin, Germany
Sep 2007 - Sep 2008	MSc Cognitive and Decision Sciences Supervisor: Prof. David Lagnado	University College London, England
Sep 2005 - July 2007	Vordiplom (eqvl. BSc)	Humboldt University, Berlin, Germany
Sep 1995 - July 2004	Abitur (eqvl. A-levels)	Scheffel Gymnasium, Bad Säckingen, Germany

Awards, Grants & Scholarships

Mar 2026	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “How to get from A to B” (\$75,000)
Dec 2025	Toyota Research Institute Human-AI Kaizen Initiative grant: “Adaptive Responsibility in Human-AI Collaboration” (\$900,000; PI)
Nov 2025	Stanford Human-Centered Artificial Intelligence (HAI) Hoffman-Yee renewal grant: “Integrating intelligence: Building shared conceptual grounding for interacting with Generative AI” (\$2,000,000; co-PI with Prof. Maneesh Agrawala (PI) and others)

Oct 2025	HAI-Google Cloud Credits Grant: “Multi-modal inference through mental simulation” (\$5,000)
Mar 2025	Bitts & Watts Initiative seed grant: “Integrating Socio-Psychological Perspectives in AI Model Selection for Performance and Sustainability” (\$100,000; co-PI with Prof. Khalid Osman (PI))
Nov 2024	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “A User-Centered Task-Driven Characterization of the Influence of AI Assistance on Clinical Decision-Making” (\$75,000; co-PI with Prof. Shriti Raj (PI), Prof. Jonathan Chen and Prof. Behnam Rahdari)
Nov 2024	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “Generalizable Physical Intuition for Robots” (\$75,000; co-PI with Prof. Jeanette Bohg (PI) and Prof. Kayvon Fatahalian)
July 2024	Stanford Human-Centered Artificial Intelligence (HAI) Hoffman-Yee grant: “Integrating intelligence: Building shared conceptual grounding for interacting with Generative AI” (\$500,000; co-PI with Prof. Maneesh Agrawala (PI) and others)
July 2024	HAI-IBM funding “Teach Yourself Preference Optimization (TYPO): Preference Optimization without Preference Labels” (\$85,000; PI)
June 2024	Stanton Prize from the Society for Philosophy and Psychology
Oct 2023	HAI-Google Funding (\$80,000; co-PI with Prof. Noah Goodman (PI))
Sep 2023	Microsoft research grant on “Accelerating Foundation Models Research” (\$20,000 in Azure cloud credits; PI)
Sep 2023	Cooperative AI grant “ACES: ACtion Explanation through counterfactual Simulation” (\$500,000; PI)
Nov 2022	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “Creative physical problem solving in humans and robots” (\$75,000; co-PI with Prof. Jeannette Bohg (PI))
Aug 2022	Stanford Human-Centered Artificial Intelligence (HAI) Hoffman-Yee grant: “MARPLE: Explaining what happened through multi-modal simulation” (\$500,000; PI)
Sep 2021	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “In touch with causation” (\$75,000; co-PI with Prof. Sean Follmer (PI))
July 2021	Templeton Foundation: Attributions of Purpose and the Philosophy of Religion (\$250,000; co-PI with Prof. Joshua Knobe (PI))
June 2021	John Philip Coghlan Fellowship from Stanford University (\$4,000)
Dec 2020	US-Israel Binational Science Foundation (BSF) “Judging responsibility under uncertainty” (\$102,600; PI)
Sep 2020	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “The Science and Engineering of Explanations (SEE)” (\$75,000; PI)
Jun 2019	Stanford Hellman Faculty Scholar Award: “Multi-modal inference through mental simulation” (\$39,960; PI)
May 2019	Stanford Human-Centered Artificial Intelligence (HAI) seed grant: “Multi-modal inference in brains, minds, and machines” (\$75,000; PI)
2015 - 2016	Templeton Foundation: “The Experience Project: Computational models of the intuitive theory of transformative experiences” (\$90,000; co-PI with Prof. Tomer Ullman (PI))
Apr 2013 - 2014	MIT Intelligence Initiative PostDoc Scholarship
May 2004 - Dec 2012	e-fellows.net Scholarship

Sep 2009 - Sep 2012	AXA PhD Scholarship (€120,000)
Apr 2012	PPG Conference Best Presentation Award, London, England <i>Award for best oral presentation at the annual conference of UCL's postgraduate peer group</i>
Mar 2011	UCL Sully Scholarship (£1,466), London, England <i>Award for best PhD upgrade talk in the Department of Cognitive, Perceptual, and Brain Sciences, 2010</i>
Jan 2011	UCL Bogue Scholarship (£3,450), London, England <i>Grant to cover 3-month lab visits to Stanford and MIT</i>
Oct 2010	KogWis, Berlin, Germany <i>Conference travel grant</i>
Feb 2006 - Sep 2009	Scholar of the German National Academic Foundation
July 2009	Max Planck Institute Berlin, Germany <i>2nd prize in poster competition of the Summer Institute on Bounded Rationality</i>
Nov 2008	University College London, England <i>Best performing student of the academic year for the MSc in Cognitive and Decision Sciences</i>
July 2006	Humboldt University, Berlin, Germany <i>Winner of the competition for the first psychological experiment</i>

Publications

Note: * indicates joint first authorship.

Submitted papers

1. Beller, A., Y. Xu, M. Siegel, S. Grant, A. Brown, J. B. Tenenbaum, S. W. Linderman, and **T. Gerstenberg** (submitted). Multimodal inference through mental simulation. <https://osf.io/preprints/psyarxiv/x2tj9>.
2. Gandhi, K., Z. Lynch, J.-P. Fränken, K. Patterson, S. Wambu, **T. Gerstenberg**, D. C. Ong, and N. D. Goodman (submitted). Human-like Affective Cognition in Foundation Models. <https://arxiv.org/abs/2409.11733>.
3. Morris, A., J. Phillips, T. Icard, J. Knobe, **T. Gerstenberg**, and F. A. Cushman (submitted). Looking back to plan ahead: Causal judgments as a sampling approximation for action effectiveness. <https://psyarxiv.com/nq53z>.
4. Rose, D., S. Zhang, S. Bridgers, H. Gweon, and **T. Gerstenberg** (submitted). Probing counterfactual thinking without counterfactual language. <https://osf.io/preprints/psyarxiv/f8daz>.
5. Shah, A. D., B. Saeed, J. B. Tenenbaum, **T. Gerstenberg**, H. Gweon, and I. Yildirim (submitted). The computational basis of how humans reason about embodied plans.
6. Shin, S., C. Hu, P. S. Muhle-Karbe, and **T. Gerstenberg** (submitted). Constructing causal stories: How mental models shape memory, prediction, and generalization. <https://osf.io/preprints/psyarxiv/6z9t7>.
7. Ma, R., S. Mueller, T. Icard, and **T. Gerstenberg** (2026). Making Good Decisions with Counterfactuals. *PsyArXiv*. <https://osf.io/preprints/psyarxiv/pwak4>.
8. Wu, S. A., S. Sridhar, and **T. Gerstenberg** (2026). Helping and hindering as counterfactual difference-making. *PsyArXiv*. <https://osf.io/preprints/psyarxiv/mb7u6>.

Journal articles

9. Chase, E., K. A. Smith, S. Follmer, and **T. Gerstenberg** (2026). Seeing, hearing, and feeling causation. *Cognitive Psychology*.
10. Harding, J., **T. Gerstenberg**, and T. Icard (2026). A Communication-First Account of Explanation. *Noûs*.
11. Beller, A. and **T. Gerstenberg** (2025). Causation, meaning, and communication. *Psychological Review*.
12. Davis, Z. J., K. R. Allen, M. Kleiman-Weiner, J. Jara-Ettinger, and **T. Gerstenberg** (2025). Inference from social evaluation. *Journal of Personality and Social Psychology*.
13. Johnson, S. G., A.-H. Karimi, Y. Bengio, N. Chater, **T. Gerstenberg**, K. Larson, S. Levine, M. Mitchell, I. Rahwan, B. Schölkopf, et al. (2025). Imagining and building wise machines: The centrality of AI metacognition. *Trends in Cognitive Sciences*.
14. Kirfel, L., **T. Gerstenberg**, and R. Zultan (2025). A framework for blaming willful ignorance. *Current Opinion in Psychology*, 102090.
15. Kirfel, L., R. J. MacCoun, T. Icard, and **T. Gerstenberg** (2025). When AI meets counterfactuals: The ethical implications of counterfactual world simulation models. *AI and Ethics*.
16. Prinzing, M., D. Rose, S. Zhang, E. Tu, A. Concha, J. Schaffer, M. Rea, **T. Gerstenberg**, and J. Knobe (2025). From artifacts to human lives: Investigating the domain-generalty of judgments about purposes. *Journal of Experimental Psychology: General*.
17. Rose, D., S. Zhang, S. Nichols, E. Markman, and **T. Gerstenberg** (2025). How children map causal verbs to different causes across development. *Nature Human Behaviour*.
18. Xiang, Y., S. J. Gershman, and **T. Gerstenberg** (2025). A signaling theory of self-handicapping. *Cognition*.
19. Amemiya, J., G. D. Heyman, and **T. Gerstenberg** (2024). Children use disagreement to infer what happened. *Cognition*.
20. **Gerstenberg, T.** (2024). Counterfactual simulation in causal cognition. *Trends in Cognitive Sciences*.
21. **Gerstenberg, T.**, D. A. Lagnado, and R. Zultan (2023). Making a positive difference: Criticality in groups. *Cognition*.
22. Gong, T., **T. Gerstenberg**, R. Mayrhofer, and N. R. Bramley (2023). Active causal structure learning in continuous time. *Cognitive Psychology* **140**, 101542.
23. Wu, S. A. and **T. Gerstenberg** (2023). If not me, then who? Responsibility and replacement. *Cognition*.
24. Zhou, L., K. A. Smith, J. B. Tenenbaum, and **T. Gerstenberg** (2023). Mental Jenga: A counterfactual simulation model of causal judgments about physical support. *Journal of Experimental Psychology: General*.
25. **Gerstenberg, T.** (2022). What would have happened? Counterfactuals, hypotheticals, and causal judgments. *Philosophical Transactions of the Royal Society B: Biological Sciences*.
26. Kirfel, L., T. F. Icard, and **T. Gerstenberg** (2022). Inference from explanation. *Journal of Experimental Psychology: General*.
27. **Gerstenberg, T.**, N. D. Goodman, D. A. Lagnado, and J. B. Tenenbaum (2021). A counterfactual simulation model of causal judgments for physical events. *Psychological Review* **128**(6), 936–975.
28. **Gerstenberg, T.** and S. Stephan (2021). A counterfactual simulation model of causation by omission. *Cognition* **216**, 104842.

29. Kominsky, J. F., **T. Gerstenberg**, M. Pelz, M. Sheskin, H. Singmann, L. Schulz, and F. C. Keil (2021). The trajectory of counterfactual simulation in development. *Developmental Psychology* **57**(2), 253–268.
30. Langenhoff, A. F., A. Wiegmann, J. Y. Halpern, J. B. Tenenbaum, and **T. Gerstenberg** (2021). Predicting responsibility judgments from dispositional inferences and causal attributions. *Cognitive Psychology* **129**, 101412.
31. Sosa, F. A., T. Ullman, J. B. Tenenbaum, S. J. Gershman, and **T. Gerstenberg** (2021). Moral dynamics: Grounding moral judgment in intuitive physics and intuitive psychology. *Cognition* **217**, 104890.
32. **Gerstenberg, T.** and T. F. Icard (2020). Expectations affect physical causation judgments. *Journal of Experimental Psychology: General* **149**(3), 599–607.
33. Grinfeld, G., D. Lagnado, **T. Gerstenberg**, J. F. Woodward, and M. Usher (2020). Causal Responsibility and Robust Causation. *Frontiers in Psychology* **11**, 1069.
34. Niemi, L., J. Hartshorne, **T. Gerstenberg**, M. Stanley, and L. Young (2020). Moral Values Reveal the Causality Implicit in Verb Meaning. *Cognitive Science* **44**(6).
35. Morris, A., J. Phillips, **T. Gerstenberg**, and F. Cushman (2019). Quantitative causal selection patterns in token causation. *PLoS ONE* **14**(8), e0219704.
36. Bramley, N. R., **T. Gerstenberg**, R. Mayrhofer, and D. A. Lagnado (2018). Time in causal structure learning. *Journal of Experimental Psychology: Learning, Memory, and Cognition* **44**(12), 1880–1910.
37. Koskuba, K., **T. Gerstenberg**, H. Gordon, D. A. Lagnado, and A. Schlottmann (2018). What's fair? How children assign reward to members of teams with differing causal structures. *Cognition* **177**, 234–248.
38. **Gerstenberg, T.**, M. F. Peterson, N. D. Goodman, D. A. Lagnado, and J. B. Tenenbaum (2017). Eye-Tracking causality. *Psychological Science* **28**(12), 1731–1744.
39. Gershman*, S. J., T. Gerstenberg*, C. L. Baker, and F. Cushman (2016). Plans, habits, and theory of mind. *PLoS ONE* **11**(9), e0162246.
40. Alicke, M. D., D. R. Mandel, D. Hilton, **T. Gerstenberg**, and D. A. Lagnado (2015). Causal conceptions in social explanation and moral evaluation: A historical tour. *Perspectives on Psychological Science* **10**(6), 790–812.
41. Kominsky, J. F., J. Phillips, **T. Gerstenberg**, D. A. Lagnado, and J. Knobe (2015). Causal superseding. *Cognition* **137**, 196–209.
42. Lagnado, D. A., **T. Gerstenberg**, and R. Zultan (2013). Causal responsibility and counterfactuals. *Cognitive Science* **47**, 1036–1073.
43. **Gerstenberg, T.** and D. A. Lagnado (2012). When contributions make a difference: Explaining order effects in responsibility attributions. *Psychonomic Bulletin & Review* **19**(4), 729–736.
44. Zultan, R., **T. Gerstenberg**, and D. A. Lagnado (2012). Finding fault: Counterfactuals and causality in group attributions. *Cognition* **125**(3), 429–440.
45. **Gerstenberg, T.** and D. A. Lagnado (2010). Spreading the blame: The allocation of responsibility amongst multiple agents. *Cognition* **115**(1), 166–171.
46. Meder, B., **T. Gerstenberg**, Y. Hagmayer, and M. R. Waldmann (2010). Observing and Intervening: Rational and Heuristic Models of Causal Decision Making. *Open Psychology Journal* **3**, 119–135.

Peer-reviewed conference proceedings articles

47. Balamurugan, N., S. Wu, A. Chun, G. Gaw, C. Eyzaguirre, and **T. Gerstenberg** (2026). Spot the ball: A benchmark for visual social inference. *Conference on Computer Vision and Pattern Recognition (CVPR)*.

48. Besch, B. and **T. Gerstenberg** (2026). Why someone asked ‘why’: Foil inference in human and llm question interpretation. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
49. Brockbank, E., N. Dee, M. O’Keeffe, W. Mahaphanit, **T. Gerstenberg**, J. E. Fan, and R. D. Hawkins (2026). Asking the right questions? What people learn about strangers in conversation. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
50. Btesh, V., S. Wu, and **T. Gerstenberg** (2026). Action understanding with presentational goals. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
51. Hu, C., **T. Gerstenberg**, and S. A. Wu (2026). Responsibility for influencing others. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
52. Lu, X., C. Bechlivanidis, and **T. Gerstenberg** (2026). Causal judgments in complex situations. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
53. Navarre, N., T. Quillien, D. Lassiter, **T. Gerstenberg**, and N. R. Bramley (2026). Explaining order effects in counterfactual reasoning. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
54. Petrova, K., **T. Gerstenberg**, N. Madibekov, and J. Gross (2026). What you see is what you regret. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
55. Sawa, K., C. Z. Cremer, **T. Gerstenberg**, S. Manohar, and M. Mazor (2026). “I would not have done that”: Outcome knowledge distorts memory for decisions made minutes ago. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
56. Teo, V., C. A. Bergey, and **T. Gerstenberg** (2026). Causal language about social interactions. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
57. Wu, S. A., S. Zhang, and **T. Gerstenberg** (2026). How to make a reasonable person. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
58. Yang, J., L. Wong, J. E. Fan, and **T. Gerstenberg** (2026). From chopped to cooked: Design and inference in physical environments. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
59. Zhou, H., B. Besch, C. M. Wu, and **T. Gerstenberg** (2026). Effective explanations support planning under uncertainty. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent. **Oral presentation.**
60. Zhu, P., C. Chang, **T. Gerstenberg**, and H. Gweon (2026). Young children use mental simulation to reason about their performance. In: *Proceedings of the 48th Annual Conference of the Cognitive Science Society*. Ed. by G. Boleda, I. Dautriche, M. Maldonado, C. Saldana, and T. Torrent.
61. Bigelow, E., Y. Xiang, **T. Gerstenberg**, T. Ullman, and S. J. Gershman (2025). Actual or counterfactual? Asymmetric responsibility attributions in language models. *NeurIPS Workshop: “CogInterp” Interpreting Cognition in Deep Learning Models*.
62. Brockbank, E., **T. Gerstenberg**, J. E. Fan, and R. D. Hawkins (2025). How do we get to know someone? Diagnostic questions for inferring personal traits. In: *Proceedings of the 47th Annual*

- Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
63. Btesh, V., D. A. Lagnado, and **T. Gerstenberg** (2025). Taking others for granted: Balancing personal and presentational goals in action selection. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley. **Oral presentation**.
 64. Harding, J., **T. Gerstenberg**, and T. Icard (2025). Toward a formal pragmatics of explanation. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 65. Parés-Morlans, C., M. Yi, C. Chen, S. A. Wu, R. Antonova, **T. Gerstenberg**, and J. Bohg (2025). Causal-PIK: Causality-based Physical Reasoning with a Physics-Informed Kernel. *International Conference on Machine Learning*.
 66. Petrova, K., J. Gross, and **T. Gerstenberg** (2025). Learning from ‘what might have been’: A Bayesian model of learning from regret. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 67. Rose, D., C. Hou, S. Nichols, **T. Gerstenberg**, and E. M. Markman (2025). Cause and fault in development. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley. **Oral presentation**.
 68. Teo, V., S. A. Wu, E. Brockbank, and **T. Gerstenberg** (2025). Leave a trace: Recursive reasoning about deceptive behavior. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 69. Wong, L., K. M. Collins, L. Ying, C. E. Zhang, **T. Gerstenberg**, T. O’Donnell, A. Lew, J. Andreas, T. Brooke-Wilson, and J. B. Tenenbaum (2025). Modeling open-world cognition as on-demand synthesis of probabilistic models. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley. **Oral presentation**.
 70. Wong, L., K. M. Collins, L. Ying, C. E. Zhang, A. Weller, **T. Gerstenberg**, T. O’Donnell, A. K. Lew, J. D. Andreas, J. B. Tenenbaum, and T. Brooke-Wilson (2025). Modeling Open-World Cognition as On-Demand Synthesis of Probabilistic Models. *NeurIPS Workshop: Bridging Language, Agent, and World Models for Reasoning and Planning*. **Spotlight**.
 71. Xiang, Y., E. Bigelow, **T. Gerstenberg**, T. D. Ullman, and S. Gershman (2025). How language models assign responsibility. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 72. Yang, J. and **T. Gerstenberg** (2025). Learning to plan from observed and counterfactual experiences. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 73. Yu, S., A. W. M. Tan, S. Zhang, P. Mao, R. Carlson, **T. Gerstenberg**, and D. Rose (2025). Generics revisited: Analyzing generalizations in children’s books and caregivers’ speech. In: *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. Ed. by D. Barner, C. Walker, A. Ruggeri, and N. Bramley.
 74. Andukuri, C., J.-P. Fränken, **T. Gerstenberg**, and N. D. Goodman (2024). STaR-GATE: Teaching Language Models to Ask Clarifying Questions. *Conference on Language Modeling*.
 75. Brockbank, E., J. Yang, M. Govil, J. E. Fan, and **T. Gerstenberg** (2024). Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation. In: *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine.
 76. Fränken, J.-P., K. Gandhi, T. Qiu, A. Khawaja, N. D. Goodman, and **T. Gerstenberg** (2024). Procedural dilemma generation for evaluating moral reasoning in humans and language models. In:

- Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine. **Oral presentation**.
77. Fränken, J.-P., E. Zelikman, R. Rafailov, K. Gandhi, **T. Gerstenberg**, and N. D. Goodman (2024). Self-supervised alignment with mutual information: Learning to follow principles without preference labels. In: *Advances in Neural Information Processing Systems*. <http://arxiv.org/abs/2404.14313>.
 78. Jin*, E., Z. Huang*, J.-P. Fränken, W. Liu, H. Cha, E. Brockbank, S. Wu, R. Zhang, J. Wu, and **T. Gerstenberg** (2024). MARPLE: A Benchmark for Long-Horizon Inference. In: *Advances in Neural Information Processing Systems*. <http://arxiv.org/abs/2410.01926>.
 79. Keshmirian, A., M. Willig, B. Hemmatian, U. Hahn, K. Kersting, and **T. Gerstenberg** (2024). Chain versus common cause: Biased causal strength judgments in humans and large language models. In: *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine.
 80. Tsirtsis, S., M. Gomez-Rodriguez, and **T. Gerstenberg** (2024). Towards a computational model of responsibility judgments in sequential human-AI collaboration. In: *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine. **Oral presentation**.
 81. Wu, S. A., E. Brockbank, H. Cha, J.-P. Fränken, E. Jin, Z. Huang, W. Liu, R. Zhang, J. Wu, and **T. Gerstenberg** (2024). Whodunnit? Inferring what happened from multimodal evidence. In: *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine.
 82. Wu, S. A., X. Ren, **T. Gerstenberg**, Y. Choi, and S. Levine (2024). Resource-rational moral judgment. In: *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Ed. by L. K. Samuelson, S. Frank, M. Toneva, A. Mackey, and E. Hazeltine.
 83. Cao, A., A. Geiger, E. Kreiss, T. Icard, and **T. Gerstenberg** (2023). A Semantics for Causing, Enabling, and Preventing Verbs Using Structural Causal Models. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
 84. Chase, E., **T. Gerstenberg**, and S. Follmer (2023). Realism of Visual, Auditory, and Haptic Cues in Phenomenal Causality. In: *IEEE World Haptics Conference (WHC)*. **Best Work-in-Progress Award**.
 85. Fränken, J.-P., A. Khawaja, K. Gandhi, J. Moore, N. D. Goodman, and **T. Gerstenberg** (2023). Off The Rails: Procedural Dilemma Generation for Moral Reasoning. In: *AI Meets Moral Philosophy and Moral Psychology Workshop (NeurIPS 2023)*.
 86. Fränken, J.-P., S. Kwok, P. Ye, K. Gandhi, D. Arumugam, J. Moore, A. Tamkin, **T. Gerstenberg**, and N. D. Goodman (2023). Social Contract AI: Aligning AI Assistants with Implicit Group Norms. In: *Socially Responsible Language Modelling Research Workshop (NeurIPS 2023)*. **Oral presentation**.
 87. Gandhi*, K., J.-P. Fränken*, **T. Gerstenberg**, and N. D. Goodman (2023). Understanding Social Reasoning in Language Models with Language Models. In: *Advances in Neural Information Processing Systems*. **Spotlight at Datasets and Benchmarks Track**.
 88. Gonzalez, B., **T. Gerstenberg**, and J. Phillips (2023). Causal Reasoning Across Agents and Objects. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
 89. Kirfel, L., X. Bunk, R. Zultan, and **T. Gerstenberg** (2023). Father, don't forgive them, for they could have known what they're doing. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong. **Oral presentation**.

90. Kirfel, L., R. J. MacCoun, T. Icard, and **T. Gerstenberg** (2023). Anticipating the risks and benefits of counterfactual world simulation models. In: *AI Meets Moral Philosophy and Moral Psychology Workshop (NeurIPS 2023)*.
91. Nam, A., C. Hughes, T. Icard, and **T. Gerstenberg** (2023). Show and tell: Learning causal structures from observations and explanations. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
92. Nie, A., Y. Zhang, A. Amdekar, C. J. Piech, T. Hashimoto, and **T. Gerstenberg** (2023). MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In: *Advances in Neural Information Processing Systems*.
93. Rose, D., S. Zhang, Q. Han, and **T. Gerstenberg** (2023). Teleology and generics. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
94. Vasconcelos, H., M. Jörke, M. Grunde-McLaughlin, **T. Gerstenberg**, M. S. Bernstein, and R. Krishna (2023). Explanations can reduce overreliance on ai systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction* 7(CSCW1), 1–38. **Best Paper Honorable Mention at CSCW 2023**.
95. Wu, S. A., S. Sridhar, and **T. Gerstenberg** (2023). A computational model of responsibility judgments from counterfactual simulations and intention inferences. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
96. Zhang*, S., J. S. She*, **T. Gerstenberg**, and D. Rose (2023). You are what you’re for: Essentialist categorization in large language models. In: *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Ed. by M. B. Goldwater, F. Anggoro, B. Hayes, and D. C. Ong.
97. Beller, A., Y. Xu, S. Linderman, and **T. Gerstenberg** (2022). Looking into the past: Eye-tracking mental simulation in physical inference. *Cognitive Science Proceedings*.
98. Outa*, J., X. J. Zhou*, H. Gweon, and **T. Gerstenberg** (2022). Stop, children what’s that sound? Multi-modal inference through mental simulation. *Cognitive Science Proceedings*. **Oral presentation**.
99. Srivastava, A. et al. (2022). Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models. *arXiv*.
100. Vodrahalli, K., R. Daneshjou, **T. Gerstenberg**, and J. Zou (2022). Do humans trust advice more if it comes from AI? an analysis of human-AI interactions. In: *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society*, pp.763–777.
101. Vodrahalli, K., **T. Gerstenberg**, and J. Y. Zou (2022). Uncalibrated models can improve human-AI collaboration. *Advances in Neural Information Processing Systems* 35, 4004–4016.
102. Wu, S., S. Sridhar, and **T. Gerstenberg** (2022). That was close! A counterfactual simulation model of causal judgments about decisions. *Cognitive Science Proceedings*.
103. Davis, Z. J., K. R. Allen, and **T. Gerstenberg** (2021). Who went fishing? Inferences from social evaluations. In: *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. **Oral presentation**.
104. Beller, A., E. Bennett, and **T. Gerstenberg** (2020). The language of causation. In: *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*. Ed. by S. Denison, M. Mack, Y. Xu, and B. C. Armstrong. Cognitive Science Society, pp.3133–3139.
105. Bridgers, S., C. Yang, **T. Gerstenberg**, and H. Gweon (2020). Whom will Granny thank? Thinking about what could have been informs children’s inferences about relative helpfulness. In: *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.

106. Kirfel, L., T. F. Icard, and **T. Gerstenberg** (2020). Learning from explanations. In: *Proceedings of the 42nd Annual Conference of the Cognitive Science Society*.
107. Kominsky, J. F., **T. Gerstenberg**, M. Pelz, H. Singmann, M. Sheskin, and F. Keil (2019). The trajectory of counterfactual simulation in development. In: *Proceedings of the 41st Annual Conference of the Cognitive Science Society*. Ed. by A. Goel, C. Seifert, and C. Freksa. Montreal, QB: Cognitive Science Society, pp.2044–2050.
108. Yildirim, I., B. Saeed, G. Bennett-Pierre, **T. Gerstenberg**, J. B. Tenenbaum, and H. Gweon (2019). Explaining intuitive difficulty judgments by modeling physical effort and risk. *Proceedings of the 41st Annual Conference of the Cognitive Science Society*. **Oral presentation**.
109. Gates, M. A., T. L. Veuthey, M. H. Tessler, K. A. Smith, **T. Gerstenberg**, L. Bayet, and J. B. Tenenbaum (2018). Tiptoeing around it: Inference from absence in potentially offensive speech. In: *Proceedings of the 40th Annual Conference of the Cognitive Science Society*.
110. Bramley, N. R., R. Mayrhofer, **T. Gerstenberg**, and D. A. Lagnado (2017). Causal learning from interventions and dynamics in continuous time. In: *Proceedings of the 39th Annual Conference of the Cognitive Science Society*. Ed. by G. Gunzelmann, A. Howes, T. Tenbrink, and E. Davelaar. Austin, TX: Cognitive Science Society, pp.150–155. **Oral presentation**.
111. **Gerstenberg, T.**, L. Zhou, K. A. Smith, and J. B. Tenenbaum (2017). Faulty towers: A hypothetical simulation model of physical support. In: *Proceedings of the 39th Annual Conference of the Cognitive Science Society*. Ed. by G. Gunzelmann, A. Howes, T. Tenbrink, and E. Davelaar. Austin, TX: Cognitive Science Society, pp.409–414. **Oral presentation**.
112. Stephan, S., P. Willemsen, and **T. Gerstenberg** (2017). Marbles in Inaction: Counterfactual Simulation and Causation by Omission. In: *Proceedings of the 39th Annual Conference of the Cognitive Science Society*. Ed. by G. Gunzelmann, A. Howes, T. Tenbrink, and E. Davelaar. Austin, TX: Cognitive Science Society, pp.1132–1137. **Oral presentation**.
113. Yildirim, I., **T. Gerstenberg**, B. Saeed, M. Toussant, and J. B. Tenenbaum (2017). Physical problem solving: Joint planning with symbolic, geometric, and dynamic constraints. In: *Proceedings of the 39th Annual Conference of the Cognitive Science Society*. Ed. by G. Gunzelmann, A. Howes, T. Tenbrink, and E. Davelaar. Austin, TX: Cognitive Science Society, pp.3584–3589.
114. Bramley, N., **T. Gerstenberg**, and J. B. Tenenbaum (2016). Natural science: Active learning in dynamic physical microworlds. In: *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Ed. by A. Papafragou, D. Grodner, D. Mirman, and J. C. Trueswell. Austin, TX: Cognitive Science Society, pp.2567–2572. **Oral presentation**.
115. **Gerstenberg, T.** and J. B. Tenenbaum (2016). Understanding “almost”: Empirical and computational studies of near misses. In: *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Ed. by A. Papafragou, D. Grodner, D. Mirman, and J. C. Trueswell. Austin, TX: Cognitive Science Society, pp.2777–2782. **Oral presentation**.
116. Niemi, L., J. Hartshorne, **T. Gerstenberg**, and L. Young (2016). Implicit measurement of motivated causal attribution. In: *Proceedings of the 38th Annual Conference of the Cognitive Science Society*. Ed. by A. Papafragou, D. Grodner, D. Mirman, and J. C. Trueswell. Austin, TX: Cognitive Science Society, pp.1745–1750. **Oral presentation**.
117. Allen, K., J. Jara-Ettinger, **T. Gerstenberg**, M. Kleiman-Weiner, and J. B. Tenenbaum (2015). Go fishing! Responsibility judgments when cooperation breaks down. In: *Proceedings of the 37th Annual Conference of the Cognitive Science Society*. Ed. by D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, T. Matlock, C. D. Jennings, and P. P. Maglio. Austin, TX: Cognitive Science Society, pp.84–89.
118. **Gerstenberg, T.**, N. D. Goodman, D. A. Lagnado, and J. B. Tenenbaum (2015). How, whether, why: Causal judgments as counterfactual contrasts. In: *Proceedings of the 37th Annual Conference*

- of the Cognitive Science Society. Ed. by D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, J. Matlock T., C. D., and P. P. Maglio. Cognitive Science Society, pp.782–787. **Oral presentation.**
119. **Gerstenberg, T.**, J. Y. Halpern, and J. B. Tenenbaum (2015). Responsibility judgments in voting scenarios. In: *Proceedings of the 37th Annual Conference of the Cognitive Science Society*. Ed. by D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, J. Matlock T., C. D., and P. P. Maglio. Austin, TX: Cognitive Science Society, pp.788–793. **Oral presentation.**
 120. Kleiman-Weiner, M., **T. Gerstenberg**, S. Levine, and J. B. Tenenbaum (2015). Inference of intention and permissibility in moral decision making. In: *Proceedings of the 37th Annual Conference of the Cognitive Science Society*. Ed. by D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, J. Matlock T., C. D., and P. P. Maglio. Austin, TX: Cognitive Science Society, pp.1123–1128. **Oral presentation.**
 121. Bramley, N., **T. Gerstenberg**, and D. A. Lagnado (2014). The order of things: Inferring causal structure from temporal patterns. In: *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Ed. by P. Bello, M. Guarini, M. McShane, and B. Scassellati. Austin, TX: Cognitive Science Society, pp.236–241. **Oral presentation.**
 122. **Gerstenberg, T.**, N. D. Goodman, D. A. Lagnado, and J. B. Tenenbaum (2014). From counterfactual simulation to causal judgment. In: *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Ed. by P. Bello, M. Guarini, M. McShane, and B. Scassellati. Austin, TX: Cognitive Science Society, pp.523–528. **Oral presentation.**
 123. **Gerstenberg, T.**, T. D. Ullman, M. Kleiman-Weiner, D. A. Lagnado, and J. B. Tenenbaum (2014). Wins above replacement: Responsibility attributions as counterfactual replacements. In: *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Ed. by P. Bello, M. Guarini, M. McShane, and B. Scassellati. Austin, TX: Cognitive Science Society, pp.2263–2268.
 124. Kominsky, J. F., J. Phillips, J. Knobe, **T. Gerstenberg**, and D. A. Lagnado (2014). Causal supersession. In: *Proceedings of the 36th Annual Conference of the Cognitive Science Society*. Ed. by P. Bello, M. Guarini, M. McShane, and B. Scassellati. Austin, TX: Cognitive Science Society, pp.761–766. **Oral presentation.**
 125. **Gerstenberg, T.**, C. Bechlivanidis, and D. A. Lagnado (2013). Back on track: Backtracking in counterfactual reasoning. In: *Proceedings of the 35th Annual Conference of the Cognitive Science Society*. Ed. by M. Knauff, M. Pauen, N. Sebanz, and I. Wachsmuth. Austin, TX: Cognitive Science Society, pp.2386–2391.
 126. **Gerstenberg, T.** and N. D. Goodman (2012). Ping Pong in Church: Productive use of concepts in human probabilistic inference. In: *Proceedings of the 34th Annual Conference of the Cognitive Science Society*. Ed. by N. Miyake, D. Peebles, and R. P. Cooper. Austin, TX: Cognitive Science Society, pp.1590–1595.
 127. **Gerstenberg, T.**, N. D. Goodman, D. A. Lagnado, and J. B. Tenenbaum (2012). Noisy Newtons: Unifying process and dependency accounts of causal attribution. In: *Proceedings of the 34th Annual Conference of the Cognitive Science Society*. Ed. by N. Miyake, D. Peebles, and R. P. Cooper. Austin, TX: Cognitive Science Society, pp.378–383. **Oral presentation.**
 128. McCoy, J., T. Ullman, A. Stuhlmüller, **T. Gerstenberg**, and J. B. Tenenbaum (2012). Why blame Bob? Probabilistic generative models, counterfactual reasoning, and blame attribution. In: *Proceedings of the 34th Annual Conference of the Cognitive Science Society*. Ed. by N. Miyake, D. Peebles, and R. P. Cooper. Austin, TX: Cognitive Science Society, pp.1996–2001.
 129. **Gerstenberg, T.**, A. Ejova, and D. A. Lagnado (2011). Blame the skilled. In: *Proceedings of the 33rd Annual Conference of the Cognitive Science Society*. Ed. by C. Carlson, C. Hölscher, and T. Shipley. Austin, TX: Cognitive Science Society, pp.720–725.
 130. **Gerstenberg, T.**, D. A. Lagnado, M. Speekenbrink, and C. Cheung (2011). Rational order effects in responsibility attributions. In: *Proceedings of the 33rd Annual Conference of the Cognitive Sci-*

ence Society. Ed. by C. Carlson, C. Hölscher, and T. Shipley. Austin, TX: Cognitive Science Society, pp.1715–1720. **Oral presentation.**

131. Schächtele, S., **T. Gerstenberg**, and D. A. Lagnado (2011). Beyond outcomes: The influence of intentions and deception. In: *Proceedings of the 33rd Annual Conference of the Cognitive Science Society*. Ed. by L. Carlson, C. Hölscher, and T. Shipley. Austin, TX: Cognitive Science Society, pp.1860–1865.
132. **Gerstenberg, T.**, D. A. Lagnado, and Y. Kareev (2010). The dice are cast: The role of intended versus actual contributions in responsibility attribution. In: *Proceedings of the 32nd Annual Conference of the Cognitive Science Society*. Ed. by S. Ohlsson and R. Catrambone. Austin, TX: Cognitive Science Society, pp.1697–1702.

Book chapters

133. Goodman, N. D., **T. Gerstenberg**, and J. B. Tenenbaum (2024). “Probabilistic programs as a unifying language of thought”. In: *Reverse-engineering the mind: The Bayesian approach to Cognitive Science*. Ed. by T. L. Griffiths, N. Chater, and J. B. Tenenbaum.
134. Smith, K. A., J. B. Hamrick, A. N. Sanborn, P. W. Battaglia, **T. Gerstenberg**, T. D. Ullman, and J. B. Tenenbaum (2024). “Probabilistic models of physical reasoning”. In: *Reverse-engineering the mind: The Bayesian approach to Cognitive Science*. Ed. by T. L. Griffiths, N. Chater, and J. B. Tenenbaum.
135. Bramley, N. R., **T. Gerstenberg**, R. Mayrhofer, and D. A. Lagnado (2019). “Intervening in time”. In: *Time and Causality Across the Sciences*. Ed. by S. Kleinberg. Cambridge University Press, pp.86–115.
136. **Gerstenberg, T.** and J. B. Tenenbaum (2017). “Intuitive Theories”. In: *Oxford Handbook of Causal Reasoning*. Ed. by M. Waldmann. Oxford University Press, pp.515–548.
137. Lagnado, D. A. and **T. Gerstenberg** (2017). “Causation in legal and moral reasoning”. In: *Oxford Handbook of Causal Reasoning*. Ed. by M. Waldmann. Oxford University Press, pp.565–602.
138. Goodman, N. D., J. B. Tenenbaum, and **T. Gerstenberg** (2015). “Concepts in a probabilistic language of thought”. In: *The Conceptual Mind: New Directions in the Study of Concepts*. Ed. by E. Margolis and S. Lawrence. MIT Press, pp.623–653.
139. Lagnado, D. A. and **T. Gerstenberg** (2015). “A difference-making framework for intuitive judgments of responsibility”. In: *Oxford Studies in Agency and Responsibility*. Ed. by D. Shoemaker. Vol. 3. Oxford University Press, pp.213–241.
140. **Gerstenberg, T.** and D. A. Lagnado (2014). “Attributing responsibility: Actual and counterfactual worlds”. In: *Oxford Studies in Experimental Philosophy*. Ed. by J. Knobe, T. Lombrozo, and S. Nichols. Vol. 1. Oxford University Press, pp.91–130.

ArXiv

141. Teo, V., R. Jain, **T. Gerstenberg**, and M. Kleiman-Weiner (2026). AI Assistants Overassist. *arXiv*. **Best Paper Award at COLM Workshop on Agent Behavior.**
142. Du, M., A. Khazatsky, **T. Gerstenberg**, and C. Finn (2024). To Err is Robotic: Rapid Value-Based Trial-and-Error during Deployment. *arXiv*. <http://arxiv.org/abs/2406.15917>.

PhD thesis & MSc thesis

143. **Gerstenberg, T.** (2013). Making a difference: Responsibility, causality, and counterfactuals. *Unpublished PhD thesis*.
144. **Gerstenberg, T.** (2009). The allocation of responsibility amongst multiple causes. *Unpublished MSc thesis*.

Conferences & Workshops

- April 2026 Cognitive Development Society (CDS) Conference, Montreal, Canada
Invited talk in symposium: *Probing counterfactual thinking without counterfactual language*
- April Workshop on “The Unfolding World: Causal & physical cognition in humans and other animals”, Cambridge, UK
Invited talk: *Causal reasoning from the cradle to the crown*
- May Phil Johnson-Laird’s 90th birthday conference, New York, USA
Invited talk: *Talking about causation*
- July Cognitive Science Conference (48th), Rio de Janeiro, Brazil
- July 2025 Cognitive Science Conference (47th), San Francisco, USA
Talk: *Cause and fault in development (presented by David Rose)*
Talk: *Modeling open-world cognition as on-demand synthesis of probabilistic models (presented by Tyler Brooke-Wilson)*
Talk: *Taking others for granted: Balancing personal and presentational goals in action selection (presented by Victor Btesh)*
Poster: *How language models assign responsibility (presented by Yang Xiang)*
Poster: *Generics revisited: Analyzing generalizations in children’s books and caregivers’ speech (presented by David Rose)*
Poster: *Toward a formal pragmatics of explanation (presented by Jacqueline Harding)*
Poster: *Learning to plan from observed and counterfactual experiences (presented by Justin Yang)*
Poster: *Learning from ‘what might have been’: A Bayesian model of learning from regret (presented by Kate Petrova)*
Poster: *How do we get to know someone? Diagnostic questions for inferring personal traits (presented by Erik Brockbank)*
Poster: *Leave a trace: Recursive reasoning about deceptive behavior (presented by Verona Teo)*
Poster: *Spot the ball: Inferring hidden information from human behavioral cues (presented by Neha Balamurugan)*
- July 2025 41st Conference on Uncertainty in Artificial Intelligence (UAI), Rio de Janeiro, Brazil
Tutorial: [Counterfactuals in Minds and Machines](#)
- June 2025 Society for Philosophy and Psychology, Cornell University, USA
Talk: *Making good decisions by inferring counterfactual probabilities from experimental and observational data*
Talk: *A signaling theory of self-handicapping (to be presented by Yang Xiang)*
Talk: *Learning to plan through observational and counterfactual simulations (to be presented by Justin Yang)*
Talk: *Cause and fault in development (to be presented by David Rose)*
Poster: *How do we get to know someone? Diagnostic questions for inferring personal traits (to be presented by Erik Brockbank)*
Poster: *Model synthesis architectures for open world cognition (to be presented by Lionel Wong)*
- February 2025 [Bellairs Workshop on Causality: Causality in the era of foundation models](#), Barbados
Invited talk
- December 2024 Workshop on “[Causality and Large Models](#)”, NeurIPS, Vancouver, Canada [[talk recording](#)]
Invited talk

- September 2024 Workshop on “[Analyzing High-dimensional Traces of Intelligent Behavior](#)”, Institute for Pure & Applied Mathematics (IPAM), Los Angeles, USA
Invited talk: Counterfactual simulation in causal cognition
- July 2024 Cognitive Science Conference (46th), Rotterdam, Holland
Invited talk in symposium: Holding others responsible: The role of counterfactual contrasts
Talk: Procedural dilemma generation for evaluating moral reasoning in humans and language models
Talk: Do as I explain: Explanations communicate optimal interventions (presented by Lara Kirfel)
Talk: Towards a computational model of responsibility judgments in sequential human-AI collaboration (presented by Stratis Tsirtsis)
Poster: Whodunnit? Inferring what happened from multimodal evidence (presented by Sarah Wu)
Poster: Resource-rational moral judgment (presented by Sarah Wu)
Poster: Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation (presented Erik Brockbank)
Poster: Chain versus common cause: Biased causal strength judgments in humans and large language models (presented Anita Keshmirian)
- June 2024 Society for Philosophy and Psychology (SPP) Conference, Purdue University, USA
Stanton Prize Address: Beyond the here and now – Counterfactual simulation in causal cognition
Talk: Do as I explain: Explanations communicate optimal interventions
Talk: Without his cookies, he’s just a monster: A counterfactual simulation model of social explanation (presented by Erik Brockbank)
Talk: Resource-rational moral judgment (presented by Sarah Wu)
Poster: Off The Rails: Procedural Dilemma Generation for Evaluating Moral Reasoning
- June 2024 Cooperative AI Workshop, Santa Cruz, USA
- June 2024 Simons Center Workshop, Berkeley, USA
Invited Speaker: Beyond the here and now: Counterfactual simulation in causal cognition
- May 2024 [Workshop on Mental Models and Learning from Observations](#), New York, USA
Invited Speaker: Causal abstraction
- March 2024 Symposium on Causation, Complexity, Cognition, and Representation for Responsible AI (virtual event organized by Google and the Santa Fe Institute)
Invited Speaker: Counterfactual simulation in causal cognition
- Dec 2023 NeurIPS 2023 Workshop, New Orleans, USA
Talk: Social Contract AI: Aligning AI Assistants with Implicit Group Norms (presented by Philipp Fränken)
Poster: Off The Rails: Procedural Dilemma Generation for Moral Reasoning (presented by Philipp Fränken)
Poster: Anticipating the risks and benefits of counterfactual world simulation models (presented by Lara Kirfel)
Poster: MoCa: Measuring human-language model alignment on causal and moral judgment tasks (presented by Allen Nie)
- Nov 2023 [Causality in Minds and Machines Workshop](#), San Francisco, USA
Talk: Counterfactual simulation in causal cognition

- July 2023 Cognitive Science Conference (45th), Sydney, Australia
Talk: *Father, don't forgive them, for they could have known what they're doing* (presented by Lara Kirfel)
Talk: *Learning what matters: Causal abstraction in human inference* (presented by Steven Shin)
Poster: *Looking into the past: Eye-tracking mental simulation in physical inference* (presented by Ari Beller)
Poster: *A computational model of responsibility judgments from counterfactual* (presented by Sarah Wu)
Poster: *Causal reasoning across agents and objects* (presented by Bryan Gonzalez)
Poster: *You are what you're for: Essentialist categorization in large language models* (presented by Siying Zhang)
Poster: *Teleology and generics* (presented by Siying Zhang)
Poster: *Show and tell: Learning causal structures from observations and explanations* (presented by Andrew Nam)
Poster: *Children use disagreement to infer what happened* (presented by Jamie Amemiya)
Poster: *A semantics for causing, enabling, and preventing verbs using structural causal models* (presented by Angela Cao)
- July 2023 [CogSci 2023 Abstractions Workshop \(virtual\)](#)
Invited Speaker: Causal abstraction
- May 2023 Heuristics and Causality in the Social Sciences, London, UK
Invited Speaker: Inferring what happened through multi-modal mental simulation
- Feb 2023 AAAI Conference on Artificial Intelligence, Washington DC, USA
Invited Speaker: [Bridge Program on Continual Causality](#)
- Dec 2022 Neural Information Processing Systems (NeurIPS), New Orleans, USA
Invited Keynote Speaker: [Workshop on Neuro Causal and Symbolic AI \(nCSI\)](#)
Invited Keynote Speaker: [Gaze meets ML](#)
Invited Keynote Speaker: [Workshop on All Things Attention: Bridging Different Perspectives on Attention](#)
Invited Keynote Speaker: [Workshop on Deep Reinforcement Learning](#)
- Sep 2022 XAI workshop, Birkbeck University, London, UK
Invited Keynote Speaker: [Varieties of explanation](#)
- Aug 2022 Computational Cognitive Neuroscience (CCN), San Francisco, USA
Poster: *Looking into the past: Eye-tracking mental simulation in physical inference*
- July 2022 Cognitive Science Conference (44th), Toronto, Canada
Talk: *Stop, children what's that sound? Multi-modal inference through mental simulation* (presented by Joseph Outa)
Poster: *Looking into the past: Eye-tracking mental simulation in physical inference* (presented by Ari Beller)
Poster: *That was close! A counterfactual simulation model of causal judgments about decisions* (presented by Sarah Wu)
Poster: *Inferences from disagreement* (presented by Jamie Amemiya)
- July 2022 ICML Workshop ["Beyond Bayes": Paths Towards Universal Reasoning Systems](#)
Talk: *MoCa: Cognitive Scaffolding for Language Models in Causal and Moral Judgment Tasks* (presented by Allen Nie)

- July 2022 Society for Philosophy and Psychology Conference (SPP), Milan, Italy
Talk: Looking into the past: Eye-tracking mental simulation in physical inference (presented by Ari Beller)
Talk: That was close! A counterfactual simulation model of causal judgments about decision
- June 2022 Robotics Science and Systems (RSS), “[Social Intelligence in Humans and Robots](#)” workshop
Talk: That was close! A counterfactual simulation model of causal judgments about decisions (presented by Sarah Wu)
- May 2022 Annual Meeting of the Association for Computational Linguistics (ACL)
Invited Keynote Speaker: [Workshop on Commonsense Representation and Reasoning \(CSRR\)](#).
- Dec 2021 NeurIPS (virtual)
Invited talk: [Workshop on Causal Inference and Machine Learning](#).
- July 2021 Cognitive Science Conference (43rd) (virtual)
Talk: Who went fishing? Inferences from social evaluations (presented by Zach Davis).
Poster: In touch with causation: Understanding the impact of kinesthetic haptics on causality (presented by Elyse Chase).
Poster: The role of counterfactual reasoning in responsibility judgments (presented by Sarah Wu).
Poster: Eye-tracking multi-modal inference (presented by Ari Beller & Yingchen Xu).
- July 2021 Invited Panelist at [ELLIS Workshop on Causethical ML](#).
- July 2021 International Conference on Machine Learning (ICML)
Invited Talk: [Workshop on Algorithmic Recourse](#).
- July 2021 IEEE World Haptics Conference
Poster: A causal feeling: How kinesthetic haptics affects causal perception (presented by Elyse D. Z. Chase). — **Winner of the Best Work-in-Progress Paper Award.**
- June 2021 Society for Philosophy and Psychology Conference (virtual)
Talk: Inferences from explanation (presented by Lara Kirfel).
Talk: The role of counterfactual reasoning in responsibility judgments (presented by Sarah Wu).
Poster: [The language of causation](#) (presented by Aaron Beller).
Poster: [Inference from social evaluations](#) (presented by Zach Davis).
Poster: [What explains causal judgments? Counterfactual versus hypothetical simulations](#).
- July 2020 Cognitive Science Conference (42nd), Toronto, Canada (virtual)
Poster: Whom will Granny thank? Thinking about what could have been informs children’s inferences about relative helpfulness (presented by Sophie Bridgers).
Poster: The language of causation (presented by Aaron Beller).
Poster: Learning from explanations (presented by Lara Kirfel).
- Apr 2020 ICLR Eighth International Conference on Learning Representations, Virtual Conference, Formerly Addis Ababa, Ethiopia
Invited Talk: [Workshop on Causal Learning for Decision Making](#).
- Sep 2019 Psychology and Economics of Causal Reasoning Conference, London, UK
Invited Talk: “Understanding ‘why’: From Counterfactual Simulations to Explanations”.
- July 2019 Cognitive Science Conference (41st), Montreal, Canada
Talk: Explaining intuitive difficulty judgments by modeling physical effort and risk (presented by Ilker Yildirim).
Poster: The trajectory of counterfactual simulation in development (presented by Jonathan Kominsky).

- July 2019 Society for Philosophy and Psychology (SPP), San Diego, California, USA
Poster: *What happened? Reconstructing the past through vision and sound.*
- Mar 2019 AAAI symposium: Beyond curve fitting – Causation, counterfactuals, and imagination-based AI, Stanford, USA
Invited Talk: *Understanding “why” – Causation, Counterfactuals, and Imagination.*
- Oct 2018 Mental Simulation Workshop, UC Merced, USA
Invited Talk: *What happened? Reconstructing the past from vision and sound.*
- July 2018 Cognitive Science Conference (40th), Madison, Wisconsin, USA
Talk: *What happened? Reconstructing the past from vision and sound.*
Poster: *Tiptoeing around it: Inference from absence in potentially offensive speech (presented by Monica Gates).*
Poster: *Moral dynamics (presented by Felix Sosa).*
- July 2018 Society for Philosophy and Psychology (SPP), Ann Arbor, Michigan, USA
Talk: *Judgments of actual causation approximate the effectiveness of interventions (presented by Adam Morris) — William James Prize for best contributed paper by a graduate student*
- May 2018 MIT, Boston, USA
Flash Talk: *What Happened? Reconstructing the Past Through Vision and Sound*
- Jan 2018 Causal pluralism: a multi-disciplinary investigation of causality in philosophy and the sciences, Cambridge, USA
Invited Talk: *A counterfactual simulation model of causal judgment.*
- July 2017 Cognitive Science Conference (39th), London, UK
Talk: *Faulty Towers: A hypothetical simulation model of physical support.*
Talk: *Causal learning from interventions and dynamics in continuous time (presented by Neil Bramley).*
Talk: *Marbles in inaction: Counterfactual simulation and causation by omission (presented by Simon Stephan).*
Poster: *Physical problem solving: Joint planning with symbolic, geometric, and dynamic constraints.*
Poster: *Eye movement-based probabilistic models for physical scene understanding (presented by Eghbal Hosseini).*
- July 2017 Morality, Language, and Thought Workshop, Paris, France
Talk: *A counterfactual simulation model of causal judgment.*
- June 2017 Society for Philosophy and Psychology (SPP), Baltimore, USA
Talk: *A counterfactual simulation model of causation by omission.*
- June 2017 Time and causality in the sciences, Hoboken, USA
Talk: *A counterfactual simulation model of causal judgment.*
- May 2017 Brains, Minds, and Machines Workshop, San Juan, Puerto Rico
Talk: *Understanding why: From counterfactual simulations to responsibility judgments.*
- Nov 2016 Society for Judgment and Decision Making, Boston, USA
Talk: *Lucky or clever? From changed expectations to responsibility judgments.*

- Aug 2016 Cognitive Science Conference (38th), Philadelphia, USA
Talk in workshop: *The role of space, time, and causality in evaluating counterfactual closeness.*
Talk: *Understanding “almost”: Empirical and computational studies of near misses.*
Talk: *Natural science: Active learning in dynamic physical microworlds (presented by Neil Bramley).*
Talk: *Implicit measurement of motivated causal attribution (presented by Laura Niemi).*
- Aug 2016 International Conference on Thinking, Providence, USA
Organized symposium: *Holding others responsible.*
Talk in contributed symposium: *From changed expectations to attributions of responsibility.*
- June 2016 Society for Philosophy and Psychology (SPP), Austin, USA
Talk in invited symposium: *A counterfactual simulation model of causal judgments.*
- Nov 2015 Society for Judgment and Decision Making (SJDM), Chicago, USA
Poster: *A counterfactual simulation model of causal judgments.*
- Nov 2015 Psychonomics, Chicago, USA
Poster: *Inference of Intention and Permissibility in Moral Judgment (presented by Max Kleiman-Weiner). Nominated for best student poster award.*
- Aug 2015 MIT, Boston, USA
Flash Talk: *Eye-tracking Causality: A Counterfactual Simulation Model of Causal Judgments*
- July 2015 Multidisciplinary University Research Initiative (MURI) meeting, Lake Arrowhead, USA
Talk: *A counterfactual simulation model of causal judgments.*
- July 2015 Cognitive Science Conference (37th), Pasadena, USA
Talk in workshop: *A counterfactual simulation model of causal judgments.*
Talk: *How, whether, why: Causal judgments as counterfactual contrasts.*
Talk: *Responsibility judgments in voting scenarios.*
Talk: *Inference of intention and permissibility in moral decision making (presented by Max Kleiman-Weiner).*
Poster: *Go fishing! Responsibility judgments when cooperation breaks down (presented by Kelsey Allen).*
- Jan 2015 Ranch Metaphysics Workshop, Arizona, USA
Invited Talk: *Intuitive theories.*
- Aug 2014 Modality Workshop Yale, New Haven, USA
Invited Talk: *From counterfactual simulation to causal judgment.*
- July 2014 Cognitive Science Conference (36th), Quebec City, Canada
Talk: *From counterfactual simulation to causal judgment.*
Poster: *Responsibility attributions as counterfactual replacements.*
Talk: *The order of things (co-author).*
Talk: *Causal supersession (co-author).*
- June 2014 Society for Philosophy and Psychology, Vancouver, Canada
Talk: *From counterfactual simulation to causal judgment.*
- Nov 2013 Roundtable on Causal Overdetermination, Urbana-Champaign, USA
Invited Talk: *Causal responsibility and counterfactuals.*
- Nov 2013 New Orleans Workshop on Agency and Responsibility, USA
Talk (joint with David Lagnado): *A difference-making framework for intuitive judgments of responsibility.*

- Aug 2013 SPUDM conference (24th), Barcelona, Spain
***Talk in symposium:** How intentions and deception influence responsibility attributions in groups.*
***Talk in symposium:** A counterfactual model of responsibility attributions in groups.*
- Aug 2013 Cognitive Science Conference (35th), Berlin, Germany
***Poster:** Back on track – Backtracking in counterfactual reasoning.*
- July 2013 London Reasoning Workshop, London, UK
***Talk:** Back on track – Backtracking in counterfactual reasoning.*
- June 2013 PostDocs Share Their Science Event, Boston, USA
***Poster:** General principles of causal attribution.*
- June 2013 Society of Philosophy and Psychology Conference, Providence, USA
- Aug 2012 Cognitive Science Conference (34th), Sapporo, Japan
***Talk:** Noisy Networks: Unifying process and dependency accounts of causal attribution.*
***Poster:** Ping Pong in Church: Productive use of concepts in human probabilistic inference.*
***Poster:** Why blame Bob? Probabilistic generative models for counterfactual reasoning and blame attribution (presented by John McCoy and Tomer Ullman).*
- July 2012 International Conference on Thinking, University College London, UK
***Invited talk in symposium:** Finding fault: Causality and Counterfactuals in Group Attributions.*
***Talk:** Noisy Networks: Unifying process and dependency accounts of causal attribution.*
- May 2012 ‘Towards Consilience’ Workshop, London Business School, UK
- May 2012 ‘Personal and Shared Intentions’ Workshop, MPI Berlin, Germany
***Invited talk:** How intentions and deception influence attributions of responsibility in groups.*
- Apr 2012 Postgraduate Peer Group Conference, Cumberland Lodge, UK
***Talk:** Noisy Newtons - People’s intuitive understanding of physics explains their causal attributions; 1st prize in the oral presentation competition.*
- Apr 2012 Tagung experimentell arbeitender Psychologen (TEAP), Mannheim, Germany
***Talk:** Noisy Newtons: Unifying process and dependency accounts of causal attribution.*
- Jan 2012 CEU Conference on Cognitive Development, Budapest, Hungary
***Invited talk in symposium:** Noisy Newtons: People’s intuitive understanding of physics explains their cause and prevention judgments.*
- Sep 2011 Experimental Philosophy Workshop, Sheffield, England
- Aug 2011 SPUDM conference (23rd), Kingston, England
***Talk:** On perceived criticality and expected effort.*
***Talk:** Rational order effects in responsibility attributions.*
- July 2011 Cognitive Science Conference (33rd), Boston, USA
***Talk:** Rational order effects in responsibility attributions.*
***Poster:** Beyond outcomes: The influence of intentions and deception.*
***Poster:** Blame the skilled.*
- July 2011 European Association of Social Psychology Conference, Stockholm, Sweden
***Talk in symposium:** Rational order effects in responsibility attributions.*

- Mar 2011 International Conference on Behavioral Decision Making, Israel
Poster: *Responsibility Attributions in Teams; 2nd prize in the conference poster competition (presented by Ro'i Zultan).*
- Oct 2010 KogWis, Potsdam, Germany
Poster: *The Allocation of Responsibility amongst Multiple Agents.*
- Sep 2010 Actual Causation Workshop, Konstanz, Germany
- Aug 2010 Cognitive Science Conference (32nd), Portland, Oregon
Poster: *The Dice are Cast: The Influence of Intended versus Actual Contributions on Responsibility Attribution.*
- Aug 2010 Mathematical Psychology Conference, Portland, Oregon
- June 2010 AXA Talent Day, Paris, France
Poster: *The Dice are Cast: The Influence of Intended versus Actual Contributions on Responsibility Attribution.*
- May 2010 Cognitive Science and Machine Learning Summer School, Sardinia, Italy
- Apr 2010 EPS/SEPEX Joint Conference, Granada, Spain
Talk: *The Dice are Cast.*
- Aug 2009 SPUDM Conference (22nd), Rovereto, Italy
Poster: *Allocation of Responsibility amongst Multiple Agents.*
- July 2009 Summer Institute, Max Planck Institute Berlin, Germany
Poster: *Allocation of Responsibility amongst Multiple Agents; 2nd prize in the poster competition.*
- June 2009 MICRAC Workshop, Toulouse, France
Talk: *Allocation of Responsibility amongst Multiple Agents.*
- July 2008 PsyPag Conference, Manchester, England
Talk: *Presentation of MSc thesis (honorable mention).*

Invited Presentations

- | | | |
|------|---|---|
| 2026 | Social Cognitive Science Seminar (virtual) | Brown University, USA |
| | Cognitive Science Colloquium | TU Darmstadt, Germany |
| | Basis.ai | New York, USA |
| 2025 | MAIDAP talk series | Microsoft Research, USA |
| | Psychology & Neuroscience Colloquium Series | Boston College, USA |
| | CogSci Colloquium | University of California, Irvine, USA |
| 2024 | Psychology Colloquium | Göttingen University, Germany |
| | Symbolic Systems Forum | Stanford University, USA |
| 2023 | Cognitive Salon | San Francisco, USA |
| | Social seminar | Department of Psychological and Brain Sciences, University of Delaware, USA |
| | Deepmind | London, UK |

	Psychology Colloquium	University of California Santa Cruz, USA
2022	Cognitive Forum	University of California, Los Angeles, USA
	Institute of Cognitive and Brain Sciences	University of California, Berkeley, USA
	Social lunch	Stanford University, USA
	Stanford Vision and Learning Lab	Stanford University, USA
	COCOA: Converging On Causal Ontology Analyses	Virtual seminar
	CEU Department of Cognitive Science colloquium	Vienna, Austria (virtual)
	Presentation at author-meets-critics symposium for Jim Woodward's new book "Causation with a human face"	Center for Philosophy of Science, Pittsburgh University, USA
	Construction of meaning workshop	Stanford University, USA
2021	Human-Centered Artificial Intelligence Seminar	Stanford University, USA
	Computation, Cognition, and Development Lab (Prof. Tomer Ullman)	Harvard University, USA
	Biological and Behavioral, Computational and Critical Data Initiative (B2C2)	Arizona State University, USA
	Computational Cognitive Science Colloquium	TU Darmstadt, Germany
	LINGUAE Seminar (Prof. Philippe Schlenker)	Institut Nicod Paris, France
	Schuck Lab (Dr. Nicolas Schuck)	Max Planck Institute Berlin, Germany
	Computation, Cognition, and Development Lab (Prof. Tomer Ullman)	Harvard University, USA
2020	Decision Making and Economic Psychology Seminar	Ben-Gurion University, Israel
	Developmental Brownbag	UC San Diego, USA
	Computational Approaches to Social Cognition Talk Series	Harvard University, USA
	Symbolic Systems Forum	Stanford University, USA
	Computational Cognitive Science Group (Prof. Neil Bramley)	Edinburgh University, UK
2019	Robotics Lunch	Stanford University, USA
	Seminar on History and Philosophy of Science	California Institute of Technology, USA
	Cognition & Language seminar	Stanford University, USA
	Mind, Brain, Computation, and Technology seminar	Stanford University, USA
2018	FriSem (Cognitive & Neuroscience seminar)	Stanford, USA
	MIT Brain and Cognitive Sciences Retreat	Newport, USA
	Visual Attention Lab	Brigham & Women's Hospital, USA

	Northeastern Cognition Area Speaker Series	Northeastern University, USA
	The Imagination and Modal Cognition Lab (Prof. Felipe De Brigard)	Duke University, USA
2017	Cognitive Lunch	Yale, USA
	Computational Social Cognition boot camp	Harvard, USA
	ConCats seminar series	NYU, USA
	Language Learning Lab (Prof. Joshua Hartshorne)	Boston College, USA
	Psychology Department Colloquium	Berkeley, USA
	Psychology Department Colloquium	Stanford University, USA
	Cushman & Greene lab meeting	Harvard University, USA
	Behavioral Science Seminar	The University of Chicago Booth School of Business, USA
2016	Computational Cognitive Neuroscience Lab (Prof. Sam Gershman)	Harvard University, USA
	Applied Statistics Workshop	Harvard University, USA
	Program for Evolutionary Dynamics	Harvard University, USA
	Center for Adaptive Rationality	Max Planck Institute Berlin, Germany
	Language and Cognition seminar series	Harvard University, USA
	Cognition seminar series	Brown University, USA
2015	Crockett lab (Prof. Molly Crockett)	Oxford University, UK
	London Judgment and Decision Making Seminar	University College London, UK
	Office of Naval Research	Washington, DC, USA
2014	Boston Moral Reasoning Group	Boston University, USA
	Social Lunch	Harvard University, Cambridge, USA
	Adaptive Behavior and Cognition Research Group (Prof. Gerd Gigerenzer)	Max Planck Institute, Berlin, Germany
	Computation and Cognition Lab (Prof. Noah Goodman)	Stanford University, USA
	Lab meeting (Prof. Joshua Knobe)	Yale University, USA
2013	Moral Cognition Lab (Prof. Joshua Greene)	Harvard University, USA
	Morality Lab (Prof. Liane Young)	Boston College, USA
	Sloman Lab (Prof. Steven Sloman)	Brown University, USA
	Early Childhood Cognition Lab (Prof. Laura Schulz)	Massachusetts Institute of Technology, USA
	Computational Cognitive Science Lab (Prof. Josh Tenenbaum)	Massachusetts Institute of Technology, USA
2012	Invited Speakers Series	University of Leicester, England

	Special Seminar (Dr. Christopher Summerfield)	Oxford University, England
	Lab meeting (Dr. Ralf Mayrhofer)	University Göttingen, Germany
2011	Computational Cognitive Science Group (Prof. Josh Tenenbaum)	Massachusetts Institute of Technology, Boston, USA
	CogLunch of the Cognitive Science Department	Massachusetts Institute of Technology, Boston, USA
	Lab meeting (Prof. Noah Goodman)	Stanford University, Palo Alto, USA
2010	Research seminar series	Decision Technology, London, England
	London Judgement and Decision Making Group	University College London, England
	Strategic Interaction group seminar (Prof. Werner Güth)	Max Planck Institute for Economics, Jena, Germany
2009	Cognitive and Decision Sciences seminar (Prof. Michael Waldmann)	University Göttingen, Germany
	Philosophy of Mind seminar (Prof. Michael Pauen)	Humboldt University, Berlin, Germany

Public outreach

We share my lab's work via [twitter](#) and [youtube](#).

1) Press

- 2025 [Cracking the code of why, when some choose to 'self-handicap'](#) – Press release in *The Harvard Gazette* on the paper “A signaling theory of self-handicapping”.
- 2024 [Humans Use Counterfactuals to Reason About Causality. Can AI?](#) – Press release on the paper “Counterfactual simulation in causal cognition”.
[Children's ability to detect ambiguity in disagreements sharpens between ages 7 and 11](#) – News article about the paper “Children use disagreement to infer what happened”.
- 2022 [‘Worse’ AI Counterintuitively Enhances Human Decision Making and Performance](#) – Press release about the paper “Uncalibrated models can improve Human-AI collaboration”.
- 2021 [Modeling How People Make Causal Judgments](#) – Press release on the paper “A counterfactual simulation model of causal judgments for physical events”.
- 2018 [Quantifying How We Intuit the Physical World](#) – Medium blog post about the *Intuitive experimentation in the physical world* paper.
- 2017 [How we determine blame](#) – Article on MIT News covering our *Eye-tracking causality* paper.
[Wo das essen nicht kalt wird](#) – Article in Deutsche Universitätszeitung (DUZ) on what working at MIT is like.
- 2012 [Are you responsible for the outcome of the election?](#) – Blog entry written by Tania Lombrozo which covers the “When contributions make a difference: Explaining order effects in responsibility attributions” paper.
- 2010 [AXA Talent Day](#) – Video created in the context of a talent day organized by AXA in Paris.

[Who is to blame when groups succeed or fail?](#) – Blog entry written by Art Markman in *Psychology Today* summarizing the “Spreading the blame: The allocation of responsibility amongst multiple agents” paper.

2) Blog Posts & Podcasts

- 2025 [How We Think About Causality and Moral Responsibility](#) – Interview on “The Dissenter” podcast.
- 2022 [The Brains Blog: “Anything goes! From a pluralism of methods towards a unified theory of causal cognition”](#)
- 2021 [Stanford Psychology Podcast on “Whose fault is it? Causal judgments in everyday life”](#)

3) Diversity and Inclusion

You can find a statement of our lab’s research values [here](#).

- 2018–2025 Participated each year in “[Paths to PhD](#)” workshop at Stanford: Orientation program for underrepresented students
- 2019–2022 Participation in Stanford’s [CSLI summer internship](#) program.
- 2021, 2023, 2024 Speaker at Annual [Future Advancers of Science and Technology \(FAST\)](#) Symposium at Stanford.
- 2017 [Tutorials at Brains, Minds, and Machines Workshop](#) in San Juan, Puerto Rico

Teaching

- 2025 [PSYCH 198: Advanced Research](#) (Psychology Undergraduate Honors Class)
[PSYCH 252: Statistical Methods for Behavioral and Social Sciences](#) (Graduate statistics course)
- 2024 [PSYCH 198: Advanced Research](#) (Psychology Undergraduate Honors Class)
[PSYCH 252: Statistical Methods for Behavioral and Social Sciences](#) (Graduate statistics course)
- 2023 No teaching (Junior Faculty Leave)
- 2022 [PSYCH 198: Advanced Research](#) (Psychology Undergraduate Honors Class)
[PSYCH 252: Statistical Methods for Behavioral and Social Sciences](#) (Graduate statistics course)
- 2021 [PSYCH 252: Statistical Methods for Behavioral and Social Sciences](#) (Graduate statistics course)
Guest lecture on “Causal judgments and inference” at Dartmouth College in Prof. Jonathan Phillip’s class
Guest lecture on “Intuitive theories” at University College London in Dr. Constantin Re-
zlescu’s class
- 2020 [PSYCH 293: What makes a good explanation? Psychological and philosophical perspectives \(co-taught with Prof. Thomas Icard\)](#) (Graduate seminar)

PSYCH 251: Experimental methods (co-taught with Prof. Michael Frank) (Graduate methods course)

PSYCH 187: Research Methods in Cognition and Development (co-taught with Prof. Hyo Gweon) (Advanced undergraduate seminar)

PSYCH 252: Statistical Methods for Behavioral and Social Sciences (Graduate statistics course)

2019 Guest lecture in Symbolic Systems 1 on “Intuitive Theories”.

PSYCH 291: Causal Cognition (Graduate seminar)

PSYCH 252: Statistical Methods for Behavioral and Social Sciences (Graduate statistics course)

2017 Tutorial on “Mental models as probabilistic programs” in class 9.66 (Computational Cognitive Science taught by Prof. Josh Tenenbaum).

Tutorial on “Mental models as probabilistic programs” at the Brains, Minds, and Machines summer school in Woods Hole. Teaching assistant for three student projects.

Tutorial on “Mental models as probabilistic programs” at the Brains, Minds, and Machines workshop in San Juan, Puerto Rico.

2016 Lecture on “Counterfactual simulation” in class 9.66 (Computational Cognitive Science taught by Prof. Josh Tenenbaum).

2015 Kaufman Teaching Certificate Program.

2012 UCL MSc lecture: Theoretical frameworks of responsibility attribution.

UCL 1st year undergraduate psychology lab class: Causal attribution.

2011 UCL 1st year undergraduate psychology lab class: Responsibility attribution.

2010 UCL teaching demonstratorship.

Mentorship

2026 Zhenze Zhang (MSc student): “Explaining how”

Ole Hennig (Visiting student): “Responsible AI”

2025 Victor Btesh (Visiting PhD student): “Presentational goals”

Neha Balamurugan (CS MSc student): “Intention inference”

Yiheng Yao (CS MSc student): “Personal vs. situational explanations”

Hanqi Zhou (Visiting PhD student): “Explaining how”

Britt Besch (Visiting student): “Explanation foil”

José Luis Nunes (Visiting student): “Personal vs. situational explanations in LLMs”

2024 Yang Xiang (Visiting PhD student): “A signaling model of self-handicapping”

Alexa Tran (Psych summer intern): “Action abstraction”

Sunny Yu (Symbolic Systems Summer intern): “Children’s books project”

Anujin Naranbaatar: (Summer fellows program) “Action abstraction”

Tuvana Soronzonbold (Summer fellows program): “Action abstraction”

- Hannah Cha: “Whodunit”
- Mishika Govil: “Action abstraction”
- Chuqi Hu (SymSys MSc student): “Causal abstraction”
- Verona Teo (Pre-doc): “Whodunit”
- 2023 Sam Kwok (Psychology summer intern): “Social contract AI”
- Patrick Ye (Symbolic Systems summer intern): “Social contract AI”
- Shruti Sridhar (Symbolic Systems summer intern): “Simulated agents”
- Cindy Xin: “Explanation valence”
- Ayesha Khawaja: “Off the rails”
- Haoran Zhao: “Pragmatic reasoning in LLMs”
- Sunny Yu: “Children’s books project”
- Philip Miao: “Children’s books project”
- Emily Jin: “Marple: Figuring out what happened through multi-modal inference”
- Hannah Cha: “Marple: Figuring out what happened through multi-modal inference”
- Adam Chun: “Spot the ball”
- 2022 Xenia Bunk (3 month visit): “Willful ignorance”
- Jeong Yeon Shin: “Explanation valence”
- Selena She (CSLI summer intern): “Are large language models teleological essentialists?”
- Siyang Zhang: “The development of counterfactual reasoning”
- Damini Kusum: “Mental rotation in people with Aphantasia”
- Steven Shin (Visiting student): “Causal abstraction”
- Shruti Sridhar (CSLI summer intern): “Simulated agents”
- 2021 Adam Huang: “A causal conjunction fallacy in intuitive physical predictions”
- Addison Jadwin: “The role of anticipated blame in action selection”
- Joseph Outa: “The development of multi-modal integration”
- Yingchen Xu: “Modeling multi-modal inference”
- Yanal Ramzi Qushair (SymSys summer intern): “Mental simulation in people with aphantasia”
- Mansi Verma (CSLI summer intern): “Using omissions as evidence for early counterfactual reasoning”
- Shruti Sridhar (Psychology summer intern): “Counterfactual simulation with agents”
- Ricky Ma: “Counterfactual inference”
- Laila Johnston: “Productive use of concepts in human probabilistic inference”
- Yixiu Zhao (Stanford Interdisciplinary Graduate Fellow): “Abstraction in human inference”
- 2020 Bryce Lynford: “Productive use of concepts in human probabilistic inference”
- Liang Zhou: “A hypothetical simulation model of causal support”

Disha Dasgupta: “Resolving ambiguities in counterfactual reasoning”

Xi Jia (Laura) Zhou: “The development of multi-modal integration”

Tina Hua: “Eye-tracking motivated reasoning”

2019 Antonia Langenhoff (3 month visit): “Predicting responsibility judgments from dispositional inferences and causal attributions”

Lara Kirfel (3 month visit): “Learning from language: Inferences about norms and causal structure from explanations”

Ross Kempner (CSLI summer intern): “Does motivation influence causal judgments via biased counterfactual simulations?”

Alan Brown (Research assistant): “Eye-tracking causality: Reconstructing the past”

Jingren Wang (Summer intern): “Hypotheticals, counterfactuals, and causal judgments”

University service

2022 - I help run a group in the psychology department that meets once a week to meditate and present drink tea together afterwards.

2025 Chairperson at thesis defense (Imran Thobani, Philosophy)

Chairperson at thesis defense (Eugy Han, Communication)

Chairperson at thesis defense (Claire Chen, Computer Science)

Chairperson at thesis defense (Kaylee Carissa Burns, Computer Science)

Chairperson at thesis defense (Brennan Shacklett, Computer Science)

SymSys concentration advisor (Gabriella Emma Fedetto)

Psychology major advisor (Christine Elizabeth Carpenter)

Psychology major advisor (Kamryn Regina Barone)

Thesis committee member (Kanishk Gandhi, Computer Science)

Thesis committee member (Violet Xiang, Psychology)

2024 Chairperson at thesis defense (Gerry Wan, Computer Science)

Dissertation defense committee member (Marianna Zhang, Psychology)

Dissertation defense committee member (Kayla Good, Psychology)

Dissertation defense committee member (Nicky Sullivan, Psychology)

Dissertation defense committee member (Andrew Nam, Psychology)

Thesis committee member (Kate Petrova, Psychology)

Thesis committee member (Peter Zhu, Psychology)

Thesis committee member (Julio Martinez, Psychology)

Thesis committee member (Lynde Folsom, Psychology)

SymSys concentration advisor (Esha Gupta)

SymSys concentration advisor (Adrian Sun)

SymSys concentration advisor (Sofia Estefania Vera Verduzco)

- SymSys concentration advisor (Elise Paula Baumgartner)
- 2023 Chairperson at thesis defense (Duligur Ibeling, Computer Science)
Thesis committe member (Nicky Sullivan, Psychology)
SymSys concentration advisor (Sam Edward Kwok)
- 2022 Undergraduate Advisor (Charles Hicks, Symbolic Systems)
Undergraduate Advisor (Addison Reese Jadwin, Symbolic Systems)
Thesis committee member (Effie Li, Psychology)
Thesis committee member (Alex Durango, Neuroscience)
Reader of first year project (Kate Petrova, Psychology)
Chairperson at thesis defense (Negin Heravi, Computer Science)
Thesis committee member (Bryan Gonzalez, Psychology, Dartmouth University)
- 2021 Chairperson at thesis defense (Omer Korat, Linguistics)
Reader of masters thesis (Jack Beasley, Symbolic Systems)
Thesis committe member (Marianna Zhang, Psychology)
Thesis committe member (Elyse Chase, Mechanical Engineering)
Psychology major advisor (Yanal Ramzi Qushair)
SymSys concentration advisor (Mia Y Bahr)
- 2020 Chairperson at thesis defense (Evan Strasnick, Computer Science)
Chairperson at thesis defense (Francesca Zaffora Blando, Philosophy)
Chairperson at thesis defense (Sebastian Schuster, Linguistics)
Chairperson at thesis defense (Reuben Cohn, Linguistics)
Reader of PhD thesis (Erin Bennett, Psychology)
Dissertation defense committe member (Sophie Bridges, Psychology)
Dissertation defense committe member (Andrew Kyle Lampinen, Psychology)
Committee member in qualifying project (Elisa Kreiss, Linguistics)
Reader of first year project (Effie Li, Psychology)
- 2019 Chairperson at thesis defense (Prerna Nadathur, Linguistics)
Reader of first year project (Andrew Nam, Psychology)

Reviews

Since	Journal / Conference / Grant	Reviews
Mar 2025	Developmental Science	
Sep 2024	Psychological Bulletin	
Apr 2023	Management Science	
Sep 2022	Open Mind: Discoveries in Cognitive Science	
June 2022	Psychological Review	

Apr 2022	Journal of Personality and Social Psychology	
May 2021	Child Development	
Oct 2020	Social Psychological and Personality Science	
Aug 2020	Nature Computational Science	
May 2019	Journal of Experimental Psychology: General	
Apr 2019	Israel Science Foundation	
Jan 2019	Panelist at NSF panel: Perception, Action, & Cognition	
Aug 2018	Trends in Cognitive Science	
May 2018	Proceedings of the National Academy of Sciences	
May 2018	KogWis	
Mar 2018	Grant referee for National Science Foundation (NSF) proposal	
Feb 2018	Psychological Science	
Mar 2017	Cognitive Psychology	
Mar 2017	Nature Human Behavior	
Dec 2016	Grant referee for the European Research Council (ERC)	
Sep 2016	Topics in Cognitive Science	
Aug 2016	Perspectives on Psychological Science	
Sep 2015	Social Psychology	
July 2015	Learning and Motivation	
May 2015	The Oxford Handbook of Causal Reasoning	
Feb 2015	Society for Philosophy and Psychology Conference	
Dec 2014	PLoS ONE	
June 2014	Journal of Experimental Psychology: Learning, Memory & Cognition	
Oct 2013	Scandinavian Journal of Psychology	
Aug 2013	Thinking and Reasoning	
Jan 2013	Journal of Cognitive Psychology	
Jan 2013	Cognitive Science	
Nov 2012	Topoi – An International Review of Philosophy	
Nov 2012	Journal of Experimental Social Psychology	
Oct 2012	Oxford Studies of Experimental Philosophy	
Feb 2012	Quarterly Journal of Experimental Psychology	
Oct 2011	Psychonomic Bulletin and Review	
June 2011	Cognition	
Jan 2011	Cognitive Science Conference Proceedings	

Editorial positions

Committees

Skills & Interests

Interests Surfing, Beachvolleyball, Chess, Cycling, Yoga, Music, Piano, Guitar, Meditation